



SUMMER INTERNSHIP PRESENTATION

AI-Based Hiring Prediction System

Applying Machine Learning & Data Science to Automate Recruitment Decisions

Student Name:	Asim Azhar
Enrollment No.:	2410031087
Program:	B.Tech CSE — 5th Semester
Organization:	LinuxWorld Informatics Pvt. Ltd.
Duration:	27th June 2026 – 31st July 2026
Mentor:	Mr. Syed Jibbran Ali, Sr. Technical Consultant

**LinuxWorld
Informatics
Pvt. Ltd.**

AI / ML / Data Science

Project Category

Reference No.

LW/JPR/2026/1285

31st July 2026

Table of Contents

Summer Internship Presentation — AI-Based Hiring Prediction System

01 Introduction & Objective

03 Problem Statement

05 Technologies & Tools Used

07 Methodology — Step by Step

09 Expected Outcomes

11 Conclusion

02 About the Organization

04 Project Objectives & Scope

06 System Architecture

08 ML Models Comparison

10 Future Scope

12 Certificates & Acknowledgements

Introduction

AI-Based Hiring Prediction System

During the Summer Internship at LinuxWorld Informatics Pvt. Ltd., a project titled "AI-Based Hiring Prediction System" was developed. It applies Data Science, Machine Learning, and Artificial Intelligence to the Recruitment and Human Resource Management domain.

What the System Does

- ▶ Assists recruiters in making data-driven hiring decisions
- ▶ Analyzes candidates: skills, certifications, education & experience
- ▶ Uses NLP (TF-IDF) to process textual candidate information
- ▶ Trains multiple ML models and selects the best performer
- ▶ Predicts hiring outcome with probability of selection

Internship at a Glance

Organization

LinuxWorld Informatics Pvt. Ltd.

Domain

AI / ML / Data Science

Duration

27 Jun – 31 Jul 2026

Mentor

Mr. Syed Jibbran Ali

Designation

Sr. Technical Consultant

Ref. No.

LW/JPR/2026/1285

About the Organization

LinuxWorld Informatics Pvt. Ltd.

LinuxWorld Informatics Pvt. Ltd. is a leading IT training and technology solutions company specializing in Data Science, Machine Learning, AI, Cloud Computing, DevOps, and Linux/Open-Source technologies. It provides hands-on, project-based internship opportunities that bridge the gap between academic learning and real-world industry requirements.

Internship Type

Summer Internship — Project Based

Project Topic

AI-Based Hiring Prediction System

Mode

Online / Virtual Internship

Key Details

- ▶ Completion Certificate Ref.: LW/JPR/2026/1285 | Date: 31st July 2026
- ▶ Internship confirmed via email on 8th May 2026 — letter submitted to college as proof
- ▶ Mentored by Mr. Syed Jibbran Ali, Sr. Technical Consultant at LinuxWorld
- ▶ Website: <https://summerinternship.in/>

Problem Statement

Challenges in Traditional Recruitment

Recruitment involves evaluating candidates using multiple academic and professional factors — education, skills, certifications, experience, projects, salary expectations, and assessment results. Manually analyzing these for a large number of candidates requires significant time and may result in inconsistent decision-making.

Key Pain Points

Time-Consuming Screening

Manual review of hundreds of resumes takes days and delays the hiring pipeline.

Inconsistent Decisions

Human bias and fatigue lead to non-uniform evaluation of candidates.

Multi-Factor Complexity

Balancing education, skills, certifications, experience and salary expectations is complex.

No Quantified Probability

Traditional processes provide no probability score to gauge likelihood of selection.

Project Objectives & Scope

What the AI-Based Hiring Prediction System Aims to Achieve

Objectives

- ▶ Develop an ML system that predicts hiring decisions from candidate data
- ▶ Preprocess datasets — cleaning, encoding, feature scaling
- ▶ Apply TF-IDF to convert textual fields into features
- ▶ Train & compare: Logistic Regression, Random Forest, SVM, KNN
- ▶ Evaluate models using accuracy and performance metrics
- ▶ Optimize best model with GridSearchCV
- ▶ Output hiring prediction with probability of selection
- ▶ Demonstrate ML in Human Resource Management

Scope of the Project

- ▶ Load and process a hiring dataset
- ▶ Explore and visualize candidate data
- ▶ Remove unnecessary data columns
- ▶ Encode categorical variables into numerics
- ▶ Process text using TF-IDF vectorization
- ▶ Scale numerical features with StandardScaler
- ▶ Train and test multiple ML models
- ▶ Compare accuracy across models
- ▶ Optimize the selected model with GridSearchCV
- ▶ Predict hiring decisions for new candidates

Technologies & Tools Used

Python Ecosystem for Data Science & Machine Learning

Python 3

Primary programming language

Pandas

Data manipulation & preprocessing

NumPy

Numerical computations & arrays

Matplotlib

Charts & data visualization

Seaborn

Statistical data visualization

Scikit-learn

ML models, preprocessing & tuning

TF-IDF Vectorizer

NLP — text-to-numeric features

StandardScaler

Feature normalization

GridSearchCV

Hyperparameter tuning

Pipeline

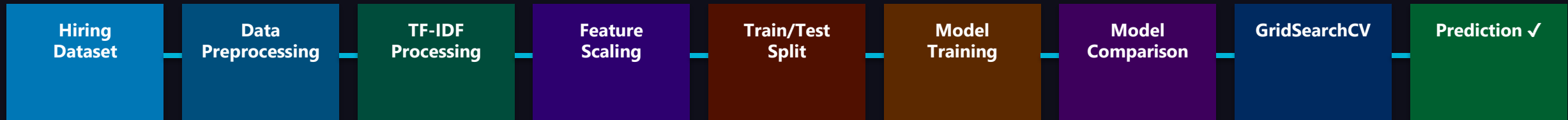
Streamlined preprocessing workflow

Google Colab

Cloud Python dev environment

System Architecture

Machine Learning Pipeline for Hiring Prediction



Main Components

1. Data Input

Load hiring dataset — academic, professional & candidate-related fields.

2. Preprocessing

Remove redundant columns; encode categorical variables to numerics.

3. Text Processing

Combine Skills, Certifications, Job Role — convert via TF-IDF.

4. Feature Scaling

Normalize numerical features with StandardScaler.

5. Model Training

Train Logistic Regression, Random Forest, SVM, and KNN classifiers.

6. Optimization & Predict

GridSearchCV tunes best model; outputs decision + probability.

Methodology

Step-by-Step Development Process

Step
1

Load Hiring Dataset

Import the hiring dataset into Python using Pandas for analysis and processing.

Step
2

Data Exploration

Understand structure, features and patterns using Matplotlib & Seaborn.

Step
3

Data Cleaning

Remove unnecessary columns; handle missing or inconsistent values.

Step
4

Encode Categoricals

Convert categorical fields to numerical representations for ML.

Step
5

Text Feature Processing

Combine Skills, Certifications, and Job Role fields for NLP processing.

Step
6

TF-IDF Vectorisation

Apply TF-IDF to convert textual information into numerical vectors.

Step
7

Feature Scaling

Normalise numerical features using StandardScaler for better training.

Step
8

Train-Test Split

Divide the processed dataset into training and testing sets.

Step
9

Model Training

Train Logistic Regression, Random Forest, SVM, and KNN models.

Step
10

Model Comparison

Compare accuracy and metrics across models to find the best one.

Step
11

Model Optimisation

Use GridSearchCV to tune hyperparameters of the best model.

Step
12

Hiring Prediction

Accept new candidate data; output decision + probability of selection.

ML Models Used & Comparison

Classifiers Trained and Evaluated

Logistic Regression

Linear classifier; interpretable baseline.

- ✓ Fast to train
- ✓ Probabilistic output
- ✓ Good baseline
- ✓ Works well linearly

Random Forest

Ensemble of decision trees; robust to noise.

- ✓ High accuracy
- ✓ Handles non-linearity
- ✓ Feature importance
- ✓ Resistant to overfitting

Support Vector Machine (SVM)

Optimal hyperplane separator; kernel-ready.

- ✓ High-dim effectiveness
- ✓ Kernel trick for NLP
- ✓ Robust to outliers
- ✓ Strong generalization

K-Nearest Neighbors (KNN)

Classifies by proximity to training points.

- ✓ Non-parametric
- ✓ Simple intuition
- ✓ No training phase
- ✓ Distance-based

The best-performing model was further optimized using GridSearchCV for hyperparameter tuning.

Expected Outcomes

What the System Delivers

Hiring Decision Prediction

Predict whether a candidate is likely to be hired.

Probability Score

Provide probability of selection alongside the binary decision.

Dual-Mode Processing

Process numerical data and textual fields in a unified pipeline.

Faster Screening

Significantly reduce time required for initial candidate screening.

Consistent Decisions

Eliminate human bias; ensure data-driven support for all decisions.

AI in HR Management

Demonstrate practical application of ML in Human Resources.

Future Scope

Planned Enhancements & Expansions

Real-Time Integration

Connect with live recruitment portals and applicant feeds.

Resume Parsing (NLP)

Auto-extract and parse candidate data from raw resumes.

Deep Learning

Adopt neural networks for improved prediction accuracy.

Web Portal

Full-stack recruiter interface for real-time predictions.

Auto Interview Scheduling

Schedule interviews for shortlisted candidates automatically.

Candidate Ranking

Rank all applicants by predicted probability of selection.

Explainable AI (XAI)

Human-readable explanations for each hiring prediction.

Cloud Deployment

Deploy on AWS / Azure for scale and enterprise access.

ATS Integration

Embed predictions into existing Applicant Tracking Systems.

Conclusion

Key Takeaways from the Internship Project

The AI-Based Hiring Prediction System demonstrates how machine learning can help recruiters make more consistent and efficient hiring choices. By combining candidate education, skills, certifications and experience with NLP-based TF-IDF text processing, the system predicts who gets hired — making screening easier and decision-making faster.

Key Achievements

- ▶ Trained & compared four ML classifiers: Logistic Regression, Random Forest, SVM, and KNN
- ▶ Implemented TF-IDF vectorization to handle complex textual candidate features
- ▶ Applied GridSearchCV to optimize the best-performing model for maximum accuracy
- ▶ Built a prediction function that outputs hiring decision + probability of selection
- ▶ Demonstrated the practical utility of AI/ML in the Human Resource Management domain
- ▶ Received formal completion certification from LinuxWorld Informatics Pvt. Ltd. (31 Jul 2026)

Project Summary

4	ML Models Trained
3	NLP Techniques Applied
35	Internship Duration(in Days)
1	Prediction Function Built

Thank You

for your time and attention

Asim Azhar

Enrollment No.: 2410031087 | B.Tech CSE, 5th Semester | IILM University

Internship: LinuxWorld Informatics Pvt. Ltd. | 27 Jun – 31 Jul 2026

Mentor: Mr. Syed Jibbran Ali, Sr. Technical Consultant

**Questions &
Feedback
Welcome**

AI-Based Hiring
Prediction System